

Predictors of the study event in a field experiment on nitrogen fertilisation in grassland plots

Abstract

Background: A field experiment on nitrogen fertilisation in grassland plots. Methods: The odds of the study event were modelled with a maximum-likelihood logistic regression (no penalisation) with nitrogen fertilisation (coded 1 for fertilised, 0 for unfertilised), age and body-mass index as predictors. Results: see the Results section. Conclusion: the findings are discussed in the context of the analysis sample.

Introduction

We report a field experiment on nitrogen fertilisation in grassland plots. The primary question was whether nitrogen fertilisation is associated with above-ground biomass (g/m^2). The dataset accompanying this report (data.csv) contains one row per plot and is described in the data dictionary.

Methods

Participants and data. Data were collected for 345 plots. Each record contains the variables listed in the data dictionary. Group membership is recorded in the column `treatment`.

Exclusion criteria and preprocessing. Only plots aged 18 to 80 years inclusive were included in the analysis. Plots with missing values for biomass, bmi were excluded (complete-case analysis).

Statistical analysis. The odds of the study event were modelled with a maximum-likelihood logistic regression (no penalisation) with nitrogen fertilisation (coded 1 for fertilised, 0 for unfertilised), age and body-mass index as predictors. Odds ratios with Wald 95% confidence intervals are reported. All analyses were performed on the analysis sample defined above. Statistics were computed with standard scientific software; no random resampling was used.

Results

Among 293 plots there were 153 events. Nitrogen fertilisation was associated with an adjusted odds ratio of 2.26 (95% CI 1.39 to 3.66, $p < 0.001$); the odds ratio per year of age was 1.02.

Discussion

The analysis followed the pre-specified plan described in the Methods. Limitations include the observational nature of some comparisons and the exclusion of records with missing values.